

Work Experience

Software Engineer, Stairwell. *Sunnyvale, CA.*

Feb 2024 - Present

- Developed gRPC/Go microservices on the Infrastructure team, primarily focused on the ingestion, indexing, scanning, and querying of a 9 billion file corpus, under strict SLOs in a cybersecurity context.
- Refactored and improved test suites to reduce flakes and decrease median time-to-merge by 5x.
- Implemented a unified data model to define relationships between files, network entities, scan results, etc. and allow nuanced graph queries with propagating tags.
- Centralized search in a query mixer, with database backends using Spanner, Bigtable, and Elasticsearch.
- Augmented our Elasticsearch indexer with rate limiting and scheduled work queues to ensure peaks of 40k attempted indexing ops/sec do not compromise search latency.
- Executed zero-downtime migrations of critical systems, including our auth service and entire 600TiB corpus.
- Triaged and resolved critical issues, built tools to reduce toil, improved monitoring, and wrote robust postmortems on an oncall rotation.
- Led the project to instrument our entire system with distributed tracing using OpenTelemetry, along with a self-hosted Grafana Tempo backend and dev tooling leading to adoption on the frontend and backend teams.

Platform Engineer, Emerald Cloud Lab. *Remote.*

June 2023 - Jan 2024

- Worked as an engineer on the Platform team, maintaining the large database and various backend services that orchestrate the cloud laboratory and support experimental development processes of biologists and chemists.
- Built an optimized REST API using Go to greatly reduce latency for common experimental procedures like HPLC.
- Designed and implemented an orchestration system for pools of Mathematica kernels hosted in AWS Fargate, allowing on-demand experimental operations using Wolfram Language to bypass minutes of startup time.
- Incorporated a custom linter for scientific Wolfram Language code as a CI check on pull requests, using GitHub webhooks and Python serverless runners on AWS Lambda.
- Fixed critical platform bugs during the 24/7 operation of the lab on an oncall rotation.

Software Developer, AI Learners. *Remote.*

Jan 2021 - Dec 2022

- Worked as a full-stack developer building a modular framework for math and logic web games using React and TypeScript, eventually supporting over 55 different games with a focus on inclusive design.
- Adopted the additional role of a project manager for one summer, planning projects and timelines, and coordinating tasks between other developers in a remote work setting.
- Ensured compatibility with assistive technologies like screen readers, adhering to WCAG standards.
- Practiced end-to-end testing with Cypress to implement robust front-end components compatible across browsers and responsive for both mobile and desktop devices.

Teaching Assistant, Cornell University Computing and Information Science. *Ithaca, NY.*

Feb 2021 - May 2023

- Facilitated lab sessions of up to 30 students, held office hours, and communicated technical concepts to students learning Java, JavaScript, HTML/CSS, game development and user-centered design.

Education

Cornell University, College of Arts and Sciences, *Ithaca, NY.*

- Master of Professional Studies in Information Science, Graduated May 2023. GPA: 4.0.
- Bachelor of Arts in Information Science, Graduated December 2022. GPA: 3.9, Dean's List.

Relevant Coursework: Advanced Game Development, Functional Programming, Object-Oriented Programming, Data Structures, Data Science, Machine Learning, Natural Language Processing, Computer Vision.

Technical Skills

- **Programming:** Proficient in Go, Python, TypeScript, Java, SQL. Experience with Rust, C#, OCaml, C++.
- **DevOps:** Proficient in Git, Prometheus, Grafana. Experience with Bazel, Docker, Temporal, Spinnaker.
- **Cloud:** Proficient in Kubernetes, Terraform, GCP (Spanner, Bigtable, PubSub, GKE, GCE). Experience with AWS.
- **Web Development:** Proficient in HTML, CSS, React, Vue. Experience with PHP.
- **Data Science:** Proficient in NumPy, pandas, R, scikit-learn. Experience with PyTorch.